

Finish the network analyser audit, EXP-004

Task	Finish the network analyser audit, EXP-004
ID	SCH-001
Class	SCHOOL-BENCH
Status	READY
Experiment	EXP-004
Decisions	0004, 0006
Last reviewed	2026-09-26
Source	docs/runbooks/SCH-001-exp004-analyser-audit.md
Source SHA-256	c8f4dfa2722e40dd19f22252d91e26ba97725957269ff4d13c26cfe9a03d5105

Why we are doing it

Five of the nine readings about the school's network analyser are incomplete: O8 was never taken, and O1, O6, O7 and O9 are only partial. Two of them unblock purchases: decision 0006 allows the first measurement enabling purchases, cables, adapters and antennas, only once O1 and O7 are recorded. This visit only looks and records. It calibrates nothing and changes nothing that has to be undone later.

The scientific question

What exactly is this analyser, what is installed on it, and can it be calibrated at the plane where our boards will connect? These are observations O1, O6, O7, O8 and O9 of EXP-004, whose frequency question is already closed by decision 0004.

What must already be true

- ☐ You have access to the school RF laboratory and permission to use the analyser.
- ☐ The analyser is the Rohde and Schwarz ZVL seen on 2026-09-20, [results/EXP-004/README.md](#) result 2.
- ☐ Nothing of ours is needed: no board, no cable, no calibration.

What you need

Item	Why it is needed
This runbook, printed or on a tablet	the observation sheet at the end is filled at the bench
A phone or camera	photographs are the evidence for O1, O7 and O8
A USB memory stick, FAT32, with an empty folder <code>AETHERARRAY</code>	the export test in step 6.6
The laptop with the repository, charged	O9, and a copy of every file before leaving
A USB A to B cable	O9; if the analyser's rear port is another type, record the type and skip O9
A pen	the sheet

Expected duration

About 30 minutes at the bench, 45 with the optional instrument check in step 6.7.

Files to bring or open

File	Where it is
This runbook	docs/runbooks/pdf/SCH-001-exp004-analyser-audit.pdf
The readings already taken, to compare with	results/EXP-004/README.md, result 2
The four documented instruments, for step 6.7	docs/hardware/measurement-bench.md, section 2.2

Files that must exist when you leave

File	What it contains
YYYY-MM-DD_EXP-004_01_rear-label.jpg	the whole rear label, legible
YYYY-MM-DD_EXP-004_06_source-level.jpg	the power field showing the level set
YYYY-MM-DD_EXP-004_07_case-N.jpg, one per case	each accessory case, open
YYYY-MM-DD_EXP-004_08_options.jpg	the options and versions page
YYYY-MM-DD_EXP-004_09_usb.png	the laptop's device list after connecting
YYYY-MM-DD_EXP-004_export-test.slp	the export test, if it could be taken
YYYY-MM-DD_EXP-004_sheet.jpg	the filled observation sheet

STOP / DO NOT CONTINUE

Stop, record why on the sheet, and leave the instrument as you found it, if:

- **the analyser is not a ZVL, or its rear label cannot be read.** Record what you see. Do not infer the model from the front panel: result 2 already did that, and it is exactly what this visit replaces.
- **someone else's setup or calibration is loaded and you have no permission to touch it.** Do not preset, and do not save over their files.
- **a reading would need a user calibration, a change of reference plane, or a cable of ours.** None of this visit needs them; they belong to a later runbook.
- **a limit can only be reached through service or factory settings.** Record the limit the normal menus accept instead.
- **the laptop asks to install a driver from an unknown source.** Do not install it; record the device name and skip the rest of O9.

Procedure

1. Open the application or instrument

1. Find the analyser and photograph the front panel as found.
2. If it is off, switch it on and wait until the measurement screen appears.
3. Do not preset it yet.

2. Load the project or set up the connection

1. There is no project to load.
2. If a setup is on screen, ask whether it may be changed. If allowed, save it under a new name, for example `AETHERARRAY_BEFORE`, so it can be restored in step 10.
3. Insert the USB stick.

3. Settings and parameters

Reading	What to do	Value to use	Source
O6, source level	set the output power field	−10 dBm	docs/hardware/measurement-bench.md section 5
O6, range	try the lowest and the highest value the field accepts	whatever it accepts	EXP-004 O6
Export test sweep	start, stop and points	2.30 to 2.60 GHz, 121 points	the EXP-011 grid

WHY

For conducted work, 0 dBm into the board with about 4 dB of chain loss would put the detector near the top of its range, so −10 dBm is the proposed default. It is far below the radiated ceiling EXP-004 derived from the regulation:

$$P_{\text{src}} \leq P_{\text{EIRP}} - G_{\text{probe}} = 10 \text{ dBm} - 2 \text{ dBi} = 8 \text{ dBm}$$

and this analyser cannot exceed 0 dBm anyway. If −10 dBm is not accepted, use the nearest accepted value and record it: O6 asks for the level actually used.

4. What not to change

- No factory or service settings, no firmware update, no option or licence change.
- No user calibration, and no file deleted or overwritten on the instrument.
- Leave the port savers on the test ports if they are fitted: they protect connectors that cost more than this project.

5. Checks before launching

- ☐ The analyser sees the USB stick: a file dialog lists it.
- ☐ Nothing is connected to either test port except port savers.
- ☐ A test photograph of small print is sharp when zoomed.

6. Launch

Take the readings in this order, and write each one on the sheet as you go.

1. **O1.** Photograph the whole rear label. Copy every field exactly, including the model, the material or order number and the serial number.
2. **O8.** Open the instrument's version and options page. On this family it is reached from the setup key, under an entry named Info, System Info or Versions and Options. Photograph it, copy every option code verbatim, and write down the menu path you used and the firmware version.
3. **O6.** Set the output power to the value in step 3, photograph the field, then try the lowest and highest accepted values and record them.
4. **O7.** Open every accessory case near the analyser. For each item record the printed model or part number, what it is (open, short, load, through, adapter, cable, torque wrench), and its connector type and sex: N, 3.5 mm or SMA, male or female. Photograph each case.
5. **O9.** Record every port on the rear panel as labelled: USB and its type, LAN, GPIB. Connect the USB cable from the analyser's device port to the laptop, open the device list, and record any new device, with its hardware identifier.

6. **Export test.** With nothing on port 1, display S11 over the sweep in step 3 and save the trace data as a Touchstone .s1p file to the USB stick. This is not a measurement of anything: it proves that data leave the instrument in a form rfkit can read.
7. **Optional.** Look for the four instruments listed in docs/hardware/measurement-bench.md section 2.2 and record each as present, with its model plate, or not seen.

7. What success looks like

- A full model string and a serial number for O1.
- A list of option codes for O8, even if it is short.
- For O7, either an inventory of the cases or the words "no case found". **No calibration kit is a valid result**, not a failure: it is the answer decision 0006 needs.
- For O9, either a device name with its identifier, or "no device appeared".
- One .s1p file on the stick whose first lines start with ! or #.

8. Export and save

1. Save a screenshot of the options page to the stick, if the analyser can print to file.
2. Copy everything from the stick to the laptop.
3. Photograph the filled sheet.

9. File names and destination

File	Name	Destination in AetherArray
Every photograph and screenshot	YYYY-MM-DD_EXP-004_<reading>_<what>.<ext>	results/EXP-004/raw/YYYY-MM-DD/
The export test	YYYY-MM-DD_EXP-004_export-test.s1p	same
The sheet	YYYY-MM-DD_EXP-004_sheet.jpg	same

raw/ stays on the laptop and is not committed; results/README.md explains why.

10. Final checklist before leaving

- ☐ The analyser is back as you found it: the saved setup recalled, or preset only if that was its state.
- ☐ The source level is no higher than you found it.
- ☐ The USB stick and cable are removed, and the cases are closed and put back.
- ☐ Every row of the sheet is filled, or marked "not obtainable" with the reason.
- ☐ The photographs are legible when zoomed.
- ☐ Every file is on both the stick and the laptop.

EVIDENCE TO BRING BACK

- The rear label photograph, O1.
- The options page photograph or screenshot, and its verbatim transcription, O8.
- The power field photograph and the accepted range, O6.
- One photograph per accessory case, and the item list, O7.
- The rear panel port list and the device list screenshot, O9.
- The .s1p export test file.
- The filled observation sheet.

BACK AT HOME

1. Put the files in `results/EXP-004/raw/YYYY-MM-DD/`, and record each one with its checksum in `results/EXP-004/SOURCE.md`: `sha256sum results/EXP-004/raw/YYYY-MM-DD/*`.
2. Check that the export test reads, from the repository root:

```
cd tools
python -c "import rfkit; print(rfkit.load_touchstone('../results/EXP-004/raw/YYYY-MM-DD/YYYY-MM-DD_EXP-004_export-test.slp', source='vna').describe())"
```

3. Fill rows O1, O6, O7, O8 and O9 of `results/EXP-004/README.md`, each checked against the data sheet of the model O1 names, and update bibliography entry T1.
4. **If O1 names a model specified below 2.44 GHz, decision 0004 reopens** by its first condition. Stop there.
5. Otherwise, with O1 and O7 recorded, decision 0006 allows the interconnect and antenna purchases, sized by what O7 found. EXP-004 closes when all nine readings are recorded.
6. Mark SCH-001 as done in `docs/runbooks/register.md`.

Observation sheet

Reading	Value	Photo or file
O1 model, material or order number, serial		
O6 level used, accepted minimum and maximum		
O7 cases and items, connector types		
O8 option codes, firmware, menu path		
O9 rear ports, device seen on the laptop		
Export test saved	yes or no	
Optional: N9923A, 8714C, N9000A, 8562A	present or not seen, each	